Chapter 28
Assessing Impacts of Climate Change on Cereal Production and Food Security in Bangladesh

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Abstract As a consequence of future climate change, agriculture is likely to be affected, which would also lead to risk of hunger and water resource scarcity with enhanced climate variability. Currently, the estimated population of Bangladesh stands at over 143.4 million and is likely to be 214.6 million in 2050. To keep pace with population growth and shrinking land resource base, the food production needs to be doubled by the year 2020. Cereal production more than doubled in the last 25 years and the production gains were achieved mainly due to yield increases.

A simulation study was conducted to assess the climate change related vulnerability of food grain production in Bangladesh. Four climate change scenarios (baseline, GFDL–TR = Geophysical Fluid Dynamics Laboratory Transient; HadCM2 = Hadley Centre Unified Model2 Transient ensemble and UKTR = UK Met. Office/Hadley

Presented at the International Conference on Global Climate Change and Its Effects, held during August 25–30, 2008 in Dhaka, Bangladesh.

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R. Lal et al. (eds.), *Climate Change and Food Security in South Asia*,
DOI 10.1007/978-90-481-9516-9_28, © Springer Science+Business Media B.V. 2011

Centre Transient) were used. Simulation runs were made for high yield varieties of rice for *Aus* (March–August), *Aman* (August–November), and *Boro* (February–July), the traditional growing seasons, using the CERES-Rice model. Simulation was carried out for wheat, which is grown from November through March, using the CERES-Wheat model. The detrimental effect of temperature rise was observed even with elevated CO_2 levels. Considerable spatial and temporal variations were also noted. Impact of these changes on food security was also assessed.

Keywords Agriculture • Bangladesh • Climate change • Food security • Rice • Wheat • CERES-Rice • CERES-Wheat • Simulation

Abbreviations

GDP	Gross Domestic Product
MoA	Ministry of Agriculture
GCM	Global Circulation Model
GFDL–TR	Geophysical Fluid Dynamics Laboratory Transient
HadCM2	Hadley Center Unified Model 2
UKTR	UK Met. Office/Hadley Center Transient
HYV	high yielding varieties
DSSAT	decision support system for agrotechnology transfer
CERES	Clouds and the earth's radiant energy system
BRRI	Bangladesh Rice Research Institute
BARI	Bangladesh Agricultural Research Institute
BARC	Bangladesh Agricultural Research Council
ICASA	International Consortium for Agricultural System Analysis
MAGICC	Model for the Assessment of the Greenhouse Gas Induced Climate Change
SCENGEN	SCENerio GENerator
HADC	Hadley Centre Unified Model 2
(HadCM2 HADC50, HADC70)	Transient Ensemble GCM Scenario

28.1 Introduction

As a consequence of future climate change agriculture is likely to be affected, which would also lead to risk of hunger and water resource scarcity with enhanced climate variability and more rapid melting of glaciers (Cruz et al. 2007). The Fourth Assessment Report of the IPCC considers agriculture as one of the most vulnerable systems to be affected by climate change in the south Asian region. Bangladesh

being one of the countries of the region, it is very likely that the climate change vulnerability would increase. Currently the share of agriculture to the total GDP is about 21.77% and the overall GDP growth rate is around 4.49%. The share of this sector is 4.9% to the total export earnings. While the share of crop sector in GDP is 12.19% (2005–2006 Fiscal Year), the contribution of fisheries has increased from 4.5% to 4.86% during the same period. Although the contribution of agriculture is not very lucrative, it employs about 63% of the labor force. Farming in the country is mainly of subsistence in nature. Therefore, livelihood of the majority of the population is directly or indirectly dependent on agriculture.

Rice is the staple of Bangladeshi people. The country produces about 6.89% of the world's paddy from 11.20 M ha of land. In the year 2006, Bangladesh produced 43.73 M t of rice paddy equivalent to 30.61 M t of clean rice. About 0.13% of the global total of wheat is also produced in the country. During 2006 the production of wheat was 0.77 M t from 0.48 M ha (FAO 2008).

The climate of Bangladesh is generally sub-tropical in the north to hot humid in the south. Southwest monsoon influences the climate during June to October, and during the winter the climate is controlled by the northeast monsoon from November to March. The summer is hot and humid and the winter is mild. The country is vulnerable to many environmental hazards, including frequent floods, droughts, cyclones, and storm surges that damage life, property, and agricultural production.

Bangladesh, a developing country in South Asia, is primarily a deltaic floodplain, and elevations in most of the country do not exceed 10 m. The country has a humid tropical climate. Rains start in March–April and end in October, and average 1,500 mm $year^{-1}$ in drier regions to 5,000 mm $year^{-1}$ in wetter regions. Winter in Bangladesh is from November to February, when the average minimum and maximum daily temperatures are about 9.7°C and 26.6°C, respectively. Summer is the hottest period of the year, with temperatures generally above 35.4°C but rarely exceeding 40.0°C. Maximum solar radiation occurs during March, April, and May and ranges from 16 to 22 MJ m^{-2} day^{-1}; minimum solar radiation occurs during December and ranges from 12 to 14 MJ m^{-2} day^{-1}. Bangladesh is highly vulnerable to severe weather events – floods and droughts are frequent and damaging. The crop most affected by floods and droughts is rice, which contributes 95.48% of the total cereal production in the country, followed by wheat (2.65%) and other cereals (1.88%) (computed from MoA 2007).

Mirza and Dixit (1997), Khan et al. (2000), and Mirza (2002) reported that during the period from 1985 to 1998 an increasing trend of about 1°C in May and 0.5°C in November has already occurred. Decadal rain anomalies above long term averages since the 1960s were also reported. In many countries of Asia, production of major cereals like rice, maize and wheat during the past few decades has declined due to increasing water stress arising partly from increasing temperature, increasing frequency of El Niño and reduction in the number of rainy days (Wijeratne 1996; Aggarwal et al. 2000; Fischer et al. 2002; Tao et al. 2003; Tao et al. 2004). Peng et al. (2004) reported in a study at the International Rice Research Institute that rice yield decreased by 10% for every 1°C increase in growing-season minimum temperature. Climate change could make it more difficult than it is already to accelerate the agricultural production to meet the growing demands in developing countries in Asia.

This study considers the possible impacts of climate change on the production of rice and wheat. It looks at the effects of three GCMs scenarios (GFDL–TR = Geophysical Fluid Dynamics Laboratory Transient; HadCM2 = Hadley Centre Unified Model2 Transient ensemble and UKTR = UK Met. Office/Hadley Centre Transient) and baseline, in six locations for rice, and in three locations for wheat. These locations represent the major rice and wheat growing regions of the country. Effort was also made to assess the impact of climate change on food security in Bangladesh.

28.2 Materials and Methods

28.2.1 Study Locations and Crops

Six sites of the country that represent the drier (Dinajpur and Jessore), wetter (Sylhet and Mymensingh), and coastal regions (Chittagong and Barisal) were chosen for the rice study, and three sites (Jessore, Mymenshing, and Dinajpur) were selected for the wheat study (Fig. 28.1). A simulation study was conducted for three high yield variety (HYV) rice crops, grown during three seasons: *Aus* (pre-monsoon-season paddy, March–August,), *Aman* (monsoon-season paddy, July–November), and *Boro* (dry-season paddy, December–May). Three widely grown varieties *BR3, BR11,* and *BRRI Dhan29* were selected for Aus, Aman, and Boro, respectively. In the simulation experiments, Aus rice was planted on 01 April, Arnan rice on 15 July, and Boro rice on 01 February. *Shourav* was the selected wheat variety which was planted on 10 November in Dinajpur, Jessore, and Mymensingh.

28.2.2 Crop Models

The simulation runs for rice and wheat were made by using CERES-Rice and CERES-Wheat crop models of Decision Support System for Agrotechnology Transfer (DSSAT) version 4.0 (Hoogenboom et al. 2003). Eighteen experiments (three crops at six locations) for rice and three experiments for wheat (one crop at three locations) were conducted. Each experiment was composed of seven treatments. Several inputs are required by these crop models. In this section, how these inputs were generated have been described.

Genetic Coefficients Both CERES models use as inputs coefficients that account for differences in growth and development among cultivars. These coefficients are referred to as genetic coefficients. Genetic coefficients allow the model to simulate performance of different varieties under diverse weather and management conditions. Therefore, to obtain reasonable outputs from the crop simulation models, it is necessary to have the appropriate genetic coefficients for the selected varieties. Since the genetic coefficients for the varieties considered in this study were not available, those were estimated following the method as described by Ritchie et al. (1998).

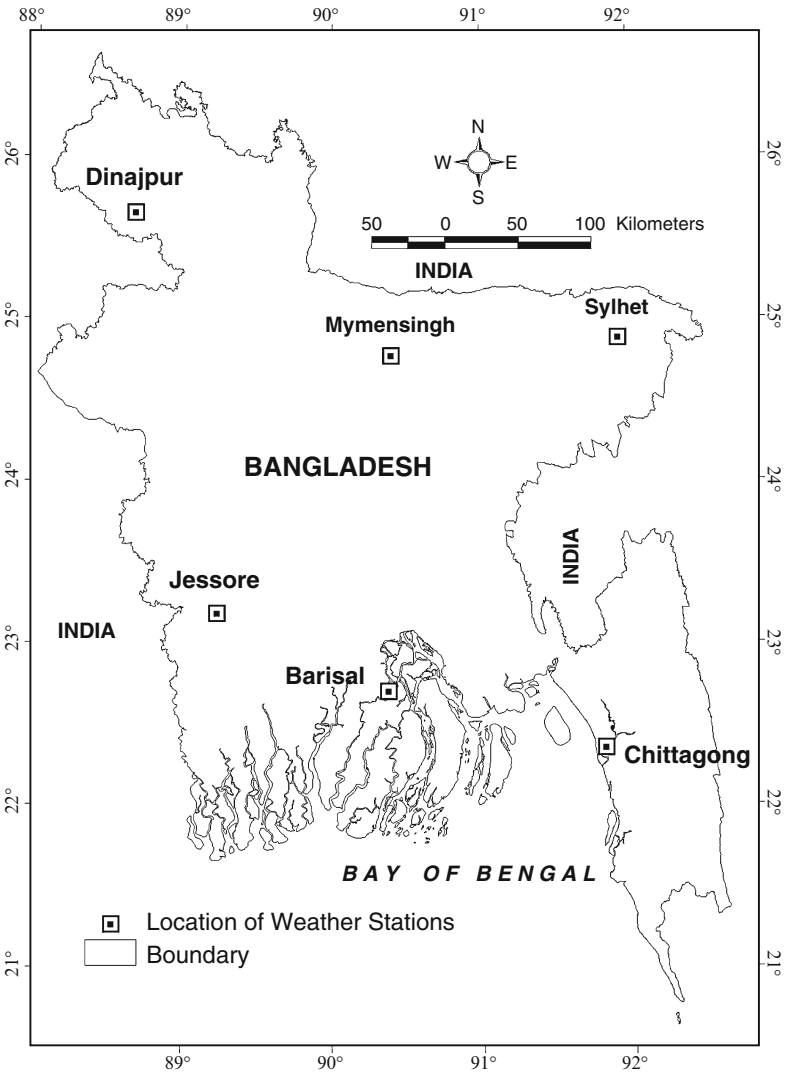


Fig. 28.1 Selected weather stations for this study

Soils Soil information is contained in a soil data file. The file contains information collected for the soil profile at a specified site along with supplemental information extracted from soil survey reports for a soil with similar taxonomic classification. Soil profiles in DSSAT v4.0 input file format were created from information compiled from various sources (Department of Soil Survey 1975) and were added to the SOILSOL file of DSSAT v4.0.

Crop Data The required crop data on development and growth characteristics and cultural practices for the selected varieties were collected from publications of

BRRI (2007) for rice, and BARI (1999) for wheat. The development and growth characteristics information was used to estimate genetic coefficients of the selected varieties. The cultural practices information was used for creating different simulation experiments.

Weather Data Twenty years of (1986–2005) daily weather data for the six sites (Fig. 28.1) was obtained from the Bangladesh Meteorological Department, which is stored at the BARC (Bangladesh Agricultural Research Council) Computer Center as ASCII files. This data set included maximum and minimum temperature (°C), rainfall (mm), and bright sunshine hour (hour) or cloud cover (Octas). Since the model requires solar radiation data (MJ m^{-2} day^{-1}), these were calculated based on the method described by Allen et al. (1998). All the weather data were then formatted as per model requirement into ICASA (International Consortium for Agricultural System Analysis) format.

28.2.3 Climate Change Scenarios

Seven scenarios were used: one baseline, and three climate change scenarios each for two intervals. The extent of change in climatic parameters or the 'delta values' of mean temperature and precipitation were generated by using MAGICC 2.4 (Model for the Assessment of Greenhouse-gas Induced Climate Change) and SCENGEN 2.4 meaning SCENario GENerator (CRU, 2000). The following model parameters for Emission Scenario IS92a (IPCC Third Assessment Report 2007) were used: According to the best guess of IPCC emission scenario IS92a, the concentrations of CO_2 for 2050 and 2070 are 515 ppmv and 575 ppmv respectively. These values were used for the simulation of crop yields. Delta values for mean temperature and precipitation were generated for the GCMs – GFDL–TR = Geophysical Fluid Dynamics Laboratory Transient, HadCM2 = Hadley Centre during 1995 and 1996 using the Second Version of the United Kingdom Meteorology Office's Unified Model UKTR = UK Met. Office/Hadley Centre Transient with respect to the base year of 1990. Default MAGGIC model parameters and high climate sensitivity were used for two time intervals, 2050 and 2070.

The range of change in mean temperature with respect to the base year (1990) was different for the three GCMs. GFDL–TR predicted delta values ranging from 1.1°C to 1.7°C for 2050, while it varied from 1.5°C to 2.3°C for 2070. Whereas, UKTR generated mean temperature increase was 1.5–2.1°C in 2050 and 1.2–2.7°C in 2070. According to HadCM2 generated delta value s, temperature would increase 1.3–2.9°C in 2050 and 1.7–4.0°C in 2070.

From Table 28.1 it can be seen that different GCMs predict different sets of values for rainfall increase (or decrease). Among the three GCMs, GFDL-TR predicted milder changes while HadCM2 suggested severe changes and the values increased with time. In case of rainfall, GFDL-TR and UKTR both predicted a decreasing tendency in the future during winter season (DJF). But, HadCM2 suggested that there would be higher precipitation (35.6% increase in 2050 and 48.9% increase in 2070) during the winter season (DJF).

Table 28.1 Delta values for mean temperature and rainfall generated for various Global Circulation Models for 2050 and 2070 using MAGICC 4.2/SCENGEN 4.2

	Global circulation models											
	GFDL–TR				HadCM2				UKTR			
	Mean temperature (°C)		Rainfall (%)		Mean temperature (°C)		Rainfall (%)		Mean temperature (°C)		Rainfall (%)	
Seasons	2050	2070	2050	2070	2050	2070	2050	2070	2050	2070	2050	2070
DJF	1.5	2.1	−1.4	−1.9	1.9	2.6	35.6	48.9	2.0	2.2	−5.2	−7.2
MAM	1.4	1.9	7.6	10.4	1.8	2.5	−1.8	−2.4	1.7	2.1	6.8	9.3
JJA	1.3	1.8	18.5	25.5	1.9	2.7	−0.2	−2.2	1.7	1.7	24.0	33.0
SON	1.3	1.8	7.4	10.3	2.1	2.9	−8.1	−5.7	1.7	1.9	−0.3	−0.4

GFDL–TR, Geophysical Fluid Dynamics Laboratory Transient; HadCM2, Hadley Centre during 1995 and 1996 using the Second Version of the United Kingdom Meteorology Office's Unified Model; UKTR, UK Met. Office/Hadley Centre Transient; DJF, December–January–February; MAM, March–April–May; JJA, June–July–August; SON, September–October–November

Table 28.2 Projected population from 1995 to 2050

Year	Population[a] (million)
1995	120.611
2000	131.155
2005	153.281
2010	166.574
2015	179.995
2020	193.129
2025	205.689
2030	217.429
2035	228.098
2040	237.575
2045	245.697
2050	252.306

[a]The World Bank HNPStats – Population Projection Tables by Country and Group.htm, URL: http://go.worldbank.org/072F5QBOC0)

28.2.4 Demography and Land Use

28.2.4.1 Population

During the period from 1961 (50.84 million) to 2001 (124.36 million) five population censuses were conducted in Bangladesh (BBS 2007)According to these census results, the population has almost doubled in less than 30 years since 1961 and is likely to be 214.6 million in 2050. Currently, the estimated population stands at over 153 million. The projections made by the World Bank (2008) show that the population would be about 205.9 million and 252.3 million under Medium Fertility Scenario and Standard Mortality Scenario in 2025 and 2050 (Table 28.2). The input data used for the projections include a base year (mid-2005) population estimate by age and sex, and base period (2005–2010) estimates of mortality, fertility, and migration (The World Bank HNPStats-Population Projection Tables by Country and Group.htm). To keep pace with the population growth and shrinking land resource base, the food production needs to be doubled by the year 2020. Sustainable growth and development of the agricultural sector is the important issue to the government for meeting the future challenge of increased production.

28.2.5 Land Use Under Major Cereals

28.2.5.1 Food Grain Production

Cereal production almost doubled in the last 25 years and the production gains were achieved mainly due to yield increases. Total production of food grains increased from 15.025 M t (million metric tons) in 1980–1981 to 27.805 M t in 2005–2006. Rice production increased by 91.10% in 2005–2006 from that of 1980–1981 (13.883 M t). On the other hand, wheat production decreased from 1.092 M t in 1980–1981 to 0.735

Table 28.3 Area and production of major cereals in Bangladesh

	2004–2005		2005–2006	
Crops	Area (M ha)	Production (M t)	Area (M ha)	Production (M t)
Local	0.574	0.638	0.517	0.664
HYV	0.451	0.862	0.517	1.081
Total Aus	1.025	1.5	1.034	1.745
Local	1.88	2.668	1.73	2.714
HYV	2.906	6.693	3.193	7.505
Total Aman	4.786	9.361	4.923	10.219
B. Aman	0.494	0.458	0.505	0.591
Total Aman	5.28	9.819	5.428	10.81
Local	0.188	0.391	0.174	0.347
HYV	3.876	13.446	3.892	13.628
Total Boro	4.064	13.837	4.066	13.975
Total rice	10.369	25.156	10.528	26.53
Wheat	0.558	0.976	0.479	0.735
Maize	0.067	0.356	0.098	0.523
Other cereals	0.038	0.018	0.027	0.017
Total cereals	11.032	26.506	11.132	27.805

Source: Bangladesh Bureau of Statistics (BBS) quoted from the Handbook of Agricultural Statistics, Ministry of Agriculture (MoA 2007) Government of Bangladesh

M t in 2005–2006 (Table 28.3). However, there was no significant change in the cultivated area. Area increased by only 2.15% for rice, while the area under wheat was reduced by about 19%. Notable increase in the area as well as the production of maize was made during the period from 1980–1981 to 2005–2006.

28.2.5.2 Food Grain Requirement

The availability of food grain in Bangladesh showed substantial improvement during the past three to 4 years. The consumption of rice, wheat, spices, edible oil, fruits, potato and fish has increased considerably, but other food items like vegetables, meat and milk consumption is still very low. The requirement of most of the energy comes from 454 g day^{-1} head^{-1} of rice and wheat. The population was 120.61 million in 1995 and 153.28 million in 2005 then the food grain requirements were 19.99 M t and 28.342 M t, respectively (Table 28.4).

28.3 Results and Discussion

28.3.1 Impact of Climate Change on Crop Yields

28.3.1.1 Rice

Under the Baseline scenario, Aus rice yields varied from a minimum of 3.08 t ha^{-1} in Jessore to a maximum of 5.08 t ha^{-1} in Dinajpur. Similarly, minimum (4.82 t ha^{-1}) and

Table 28.4 Food grain and production requirements from 1995 to 2050

Year	Food grain requirement[a] (M t)	Production requirement[b] (M t)
1995	19.986	22.301
2000	21.734	24.250
2005	25.400	28.342
2010	27.603	30.799
2015	29.827	33.281
2020	32.003	35.709
2025	34.085	38.032
2030	36.030	40.202
2035	37.798	42.175
2040	39.369	43.927
2045	40.714	45.429
2050	41.810	46.651

[a]Food grain requirement is calculated @ 454 g day^{-1} $head^{-1}$

[b]Production requirement is calculated by adding 11.58% for seed, feed and wastage with food grain requirement as per the study on 'Seed, Feed and Post Harvest losses' Ministry of Food (MOF), Government of Bangladesh

maximum (6.01 t ha^{-1}) of Aman rice yields were obtained for Barisal and Dinajpur. Maximum (6.58 t ha^{-1}) and minimum (4.66 t ha^{-1}) Boro rice yields were recorded in Sylhet and Jessore, respectively. The percentages of yield difference as compared to baseline are presented for Aus, Aman, and Boro rice at the six study sites in Table 28.5.

Aus Rice With different climate change scenarios it was evident that in the case of Aus rice under GFDL50 scenario compared to Baseline 3.8% and 15.7% yield increases were noted in Jessore and Barisal, respectively. Under GFDL70 scenario yield increases were recorded for all six locations, which varied from 3.2% in Dinajpur to 14.8% in Jessore. It is evident from Table 28.5 that Aus rice under HADC50 and HADC70 scenarios yield reduction occurred in all locations and the percentages varied from 13.5 in Jessore to 43.6 in Dinajpur. In most cases the yield reductions were around 25%. For Aus grown under UKTR50 scenario, the yield differences compared to baseline varied between +3.4% in Jessore and −15.4% in Mymensingh and Chittagong.

Aman Rice For Aman rice under GFDL50 scenario, the percent change in yields over Baseline varied between −4.9 and 1.4 in Dinajpur and Barisal, respectively. Similarly, under GFDL70 scenario the percent change in yields over Baseline varied between −5.2 and 0.1 in Dinajpur and Barisal, respectively. In the case of HADC50 and HADC70 scenarios, yield reductions were more for the latter scenario than the former. For HADC50 scenario the yield differences were between −13.5% and +1.9% in Jessore and Barisal, respectively. Yield reductions were observed in all locations under HADC50 scenario, maximum for Dinajpur (−7.7%) and minimum (−1.4%) in Jessore. The effect of UKTR50 and UKTR70 on Aman rice yields was negative in all six locations. The effect of the latter scenario was harsher than the former. Yield reductions ranged from 3.2% in Jessore to 14.2% in Dinajpur.

Table 28.5 Rice and wheat yields under different climate change scenarios

	Sylhet			Mymensingh			Dinajpur		
Simulation	Aus	Aman	Boro	Aus	Aman	Boro	Aus	Aman	Boro
Baseline (t ha^{-1})	498.9	5,439.5	6,581.5	4,169.0	5,425.9	5,664.8	5,080.3	6,006.8	5,771.8
Percent change over baseline									
GFDL50	−6.1	−1.7	6.1	−7.8	0.0	7.3	−7.0	−4.9	1.1
GFDL70	6.7	−1.7	10.7	6.1	−2.3	15.5	3.2	−5.2	9.7
HADC50	−25.7	−1.0	3.6	−26.5	−0.9	7.5	−43.6	−13.5	−5.7
HADC70	−24.0	−3.8	5.8	−25.8	−4.6	11.0	−25.8	−7.7	10.0
UKTR50	−13.4	−5.2	2.2	−15.4	−6.4	5.7	−14.1	−9.2	1.6
UKTR70	−3.9	−9.8	4.4	−6.7	−11.7	11.2	−6.3	−14.2	9.3
	Jessore			Chittagong			Barisal		
Simulation	Aus	Aman	Boro	Aus	Aman	Boro	Aus	Aman	Boro
Baseline (t ha^{-1})	3,077.3	5,115.2	4,658.7	4,169.0	5,968.9	6,059.0	3,450.6	4,819.2	6,019.3
Percent change over baseline									
GFDL50	15.7	3.3	2.5	−7.8	−0.2	0.2	3.8	1.4	3.0
GFDL70	14.8	0.6	9.3	6.1	−0.4	4.7	9.9	0.1	7.3
HADC50	−13.5	1.9	1.3	−26.5	−1.6	−0.9	−19.0	0.1	1.3
HADC70	−24.9	−1.4	6.6	−25.8	−2.6	1.0	−24.1	−1.7	3.3
UKTR50	3.4	−3.2	−0.6	−15.4	−4.6	−3.6	−5.7	−3.4	−0.6
UKTR70	−2.4	−9.0	4.7	−6.7	−7.9	−1.7	−4.3	−7.3	1.3

Table 28.6 Wheat yields under different climate change scenarios

Simulation	Dinajpur	Mymensingh	Jessore
Baseline (t ha^{-1})	3.01	2.67	2.72
Percent change over baseline			
GFDL50	20.4	28.6	29.0
GFDL70	−18.4	−15.8	−17.3
HADC50	11.2	23.8	23.0
HADC70	−23.1	−23.7	−25.6
UKTR50	17.0	24.8	24.0
UKTR70	−22.8	−21.5	−24.0

Boro Rice In general, it was noted from Table 28.5 that there is a beneficial effect of climate change in the case of Boro rice. Significant yield increase of Boro were noted for GFDL70 scenario, irrespective of locations. Under this scenario, highest yield increase was recorded in Mymensingh (15.5%), followed by Sylhet (10.7%). However, only in a few cases was a negative effect seen; −5.7% in Dinajpur for HADC50 scenario, in Chittagong −3.6 % for UKTR50, −1.7% for UKTR70 and −0.9% HADC50; and −0.6% for UKTR50 in Jessore and Barisal.

Wheat Under the Baseline scenario wheat yields varied from a minimum of 2.67 t ha^{-1} in Mymensingh to a maximum of 3.01 t ha^{-1} in Dinajpur. Simulation results in Table 28.6 show that all the scenarios for 2050 had a positive effect on wheat yield. Highest relative yield increases were around 29.0% in Jessore and Mymensingh and 20.4% in Dinajpur. Conversely, all the scenarios for 2070 had a negative effect on wheat yield. Highest relative yield decreases were 25.6% in Jessore and 23.0% in Dinajpur and Mymensingh. The reductions ranged from 15.8% to 25.6%.

28.3.2 Effect of Climate Change on Aggregated Rice and Wheat Production

Production of rice and wheat for the entire country was aggregated by crop and by season (Table 28.7). The observations are discussed in the following sections.

Aus Rice The average aggregated production of Aus rice in Bangladesh was 1.72 M t during 2005–2006. Productions were reduced by 1.5–25.8% under different climatic scenarios. Only with GFDL70 scenario was a 7.8% increase in production was recorded.

Aman Rice The aggregated production of Aman rice for 2005–2006 was 10.81 M t. Simulation results show that under all scenarios, production was lowered from 0.4% for GFDL50 scenario to 10.0% for UKTR70.

Boro Rice The baseline production of Boro rice for 2005–2006 was 13.98 M t. Under all climate change scenarios, production increased in the range of 1.2% for HADC50 and 9.5% for GFDL70 scenarios.

Table 28.7 Aggregated rice and wheat production of Bangladesh under different climate change scenarios

Simulation	Aus (M t)	Change (%)	Aman (M t)	Change (%)	Boro (M t)	Change (%)	Wheat (M t)	Change (%)	Total (M t)
Baseline	1.75	–	10.81	–	13.98	–	0.74	–	27.27
GFDL50	1.72	−1.5	10.77	−0.4	14.45	3.4	0.93	25.98	27.86
GFDL70	1.88	7.8	10.65	−1.5	15.31	9.5	0.61	−17.19	28.44
HADC50	1.30	−25.8	10.54	−2.5	14.14	1.2	0.88	19.35	26.86
HADC70	1.31	−25.1	10.42	−3.7	14.85	6.3	0.56	−24.14	27.13
UKTR50	1.57	−10.1	10.24	−5.3	14.08	0.8	0.90	21.94	26.78
UKTR70	1.66	−5.1	9.73	−10.0	14.65	4.9	0.57	−22.79	26.61

Wheat The aggregated production of wheat for 2005–2006 was 0.735 M t. It is interesting to note that with all the scenarios for 2050, the wheat production increased in the tune of 19.35–25.98%. While, under scenarios for 2070 the production decrease ranged from 17.19% for GFDL70 to 24.14% for HADC70 scenarios.

28.3.3 Effect of Climate Change on Crop Growing Season

The growing period for Aus, Aman, and Boro rice and wheat under Baseline conditions were 130, 145, and 165 and 110 days, respectively. In general, all the climate change scenarios enhanced the crop growing season by 2–10 days compared to baseline. Under all scenarios for 2070, the growing season was reduced by 8–10 days and for 2050 the reduction was 5–7 days.

Yield reductions of rice and wheat are attributed to the effect of high temperature during Aus and Aman seasons and dry spells. During Boro season the temperature increase under all scenarios varied between 1.5°C for GFDL50 scenario and 2.6°C for HADC70. These temperature increases did not exceed the threshold value of 35.0°C for rice. Wheat is sensitive to high temperature, but increase of wheat yields was observed. This may be attributed to the fact that when the temperature regime is not detrimental and wheat being a C3 crop, it is likely that at atmospheric CO_2 concentrations of 515 and 575 ppmv CO_2 fertilization might have occurred.

The existing literature suggests similar findings: Reilly et al. (1996) reported that in Asia, including Bangladesh, under different GCM scenarios and sites, the yields of cereals varied with CO_2 effect. The range of variations for rice, maize, and wheat were −22 to +28, −65 to −10, and −61 to +67, respectively. Squire and Unsworth (1988) indicated that determinate crops with discrete elements to their life cycle develop faster in higher temperatures, and so the stage of seed filling is shortened, limiting the benefits of elevated CO_2. When CO_2 concentration doubled, but daily weather data from a typical year were used, the potential wheat yield was 27% larger than in a control run at 340 m mol/mol CO_2 (11.5 and 9.0 t ha^{-1}, respectively). The date of maturity was unchanged. When daily temperatures were increased by 3°C and CO_2 was doubled, more rapid development of the crop shortened the growing season, and the potential grain yield (10.4 t ha^{-1}) was only 15% larger than in the control, but maturity occurred 30 days earlier.

Simulation studies conducted by Karim et al. (1996, 1999) showed that rice and wheat production in Bangladesh decreased due to rise in temperature. Wheat was more susceptible than rice. It was found that a 4°C increase in temperature had severe impact on food grain production, especially on wheat production. On the other hand, CO_2 fertilization would facilitate the food grain production. A rise in temperature will cause significant decrease in production, some 28% and 68% for rice and wheat, respectively. The apparent increase in yield of Boro and other crops might be constrained by moisture stress. A 60% moisture stress on top of other effects might cause as high as 32% decline in Boro yield instead of having an overall 20% net increase.

28.3.4 Food Security

Bangladesh is highly vulnerable to floods and droughts. Droughts of different intensities occur, especially in the dried regions of the country, due to uneven distribution of rainfall and a higher evapotranspiration rate accelerated by increased temperature, low humidity, and high wind speed. Wet season drought severely affects transplanted Aman rice, reducing the annual production by about 1.5 M t of rice. On the other hand, dry season drought affects the production of wheat, potato and Aus paddy (Karim et al. 1990). Drought related vulnerability can be minimized by supplemental irrigation. Attempts should be made to develop drought resistant varieties of crops for drought prone areas of the country.

One of the obvious effects of climate change may be changes in pests and weeds in agriculture. This is because this change will likely affect the pest–weed–host relationship in one (or more) of three ways: by affecting the pest–weed population; by affecting the host population; and by affecting the pest–weed–host interaction. These would make food security more vulnerable.

If the production of major cereals is hampered due to climate change then food security would be in jeopardy, as out of a recommended 2122 calories day^{-1} $person^{-1}$, the most part is met from intake of 454 g of food grains (rice and wheat). Most of the estimates show that a huge amount of additional food grain will be required to feed the growing population of the country. Hossain and Shahabuddin, (1999) estimated that at a 4.5% national income growth per annum, cereal requirement will be 30.57 and 35.39 M t in 2010 and 2020, respectively. At a 6.5% national income growth per annum cereal requirement in 2010 and 2020 would be 32.82 and 35.14 M t, respectively.

World Bank's estimate based on the assumption that NRR (net reproductive rate) would be lower from present level of about 1.2 to unity by 2025–2030, indicates a population of 131 million, 167 million and 217 million during 2000, 2010 and 2030 respectively.

Currently, the estimated population stands at over 153 million and by the year 2030 and 2050 it is projected to be 217 and 252 million, respectively (World Bank 2008 To feed this huge population, the food grain requirement of the country would be 36.03 M t in the year 2030 and 41.81 M t in the year 2050.

Therefore, to become self-sufficient in food grain production in the years 2030 and 2050, compared to the production of 2005–2006 (27.805 M t), an additional 8.22 and 14.00 M t respectively would be required. Besides, as per estimation by the Ministry of Food of Bangladesh about 11.58% additional food grain would be required for seed, feed and wastage. Therefore, the production requirement would be 40.2 M t in 2030 and 46.65 M t in 2050. Although food grain production is an important part of food security, it is not the only strategy. With the income growth the preference for food is shifted from carbohydates to more protein, vegetables and fruits. This will moreover reduce the pressure on cereals in the future.

28.4 Conclusion

Under the business as usual scenario, with the available technologies Bangladesh has the potential to meet demand for cereals up to 2025 and even beyond. However, it is difficult to predict the consequences of climate change on agriculture as a whole. Cereal demand will be difficult to meet as the population is very likely to be more than 252 million by the year 2050. However, under changed scenarios, new technologies need to be developed to combat climate change and sea level rise, coupled with the great pressure of population increase.

To predict the consequences of climate change on agriculture, limited efforts have been made to generate climate change scenarios using various Global Climate Models (GCMs) and DSSAT (Decision Support System for Agrotechnology Transfer) based Crop Simulation models of International Consortium for Agricultural Systems Applications (ICASA). Due to unavailability of high resolution GCMs for Bangladesh as well as for the South Asia region, reliable scenarios cannot be generated. On the other hand, for running different crop models, various required minimum datasets for crops, management, and climatic parameters are not collected adequately or are incomplete. Another aspect of these efforts is validation of the output from these models, which is very important. To make reliable predictions the model outputs need to be thoroughly substantiated with evidence. National facility should be developed to conduct phytotron (plant growth chamber) experiments under simulated climate change scenarios.

Since climate change and sea level rise will have adverse effect on the ecosystems, many important germplasms are likely to be lost. Research should be initiated to preserve the vulnerable species of crop, livestock, etc. Genetic tolerance to adverse environment/improvement must be achieved in order to have sustainable food security and development. Research should be undertaken for screening of genotypes for withstanding adverse edaphic, environmental, and climatic conditions. As a consequence of climate change, drier regions would become more arid in the winter and the possibility of growing rainfed crops would be diminished. Therefore, low moisture consuming crops/cropping patterns need to be developed and disseminated for wider adoption by the farmers.

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